PERSCHOLAS GREATER BOSTON

CYBERSECURITY PROGRAM

NETWORK DEFENSE MODULE - LESSON 2 LAB EXERCISES

Classwork Activity 1— VLANs in Action: Can They Talk?

Cisco Packet Tracer — Hands-On Exercise

Objective

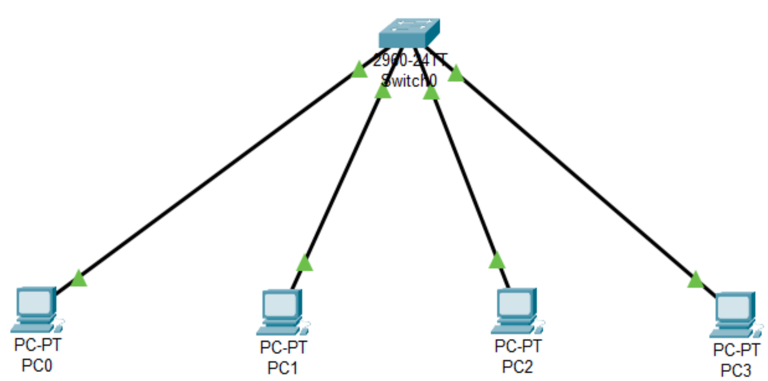
By the end of this activity, you will be able to:

- Create VLANs on a switch
- Assign ports to VLANs
- Test communication between devices in the same and different VLANs

Your Topology

You have the following devices already connected:

Device	Connected To	Port
PC0	Switch0	Fa0/1
PC1	Switch0	Fa0/2
PC2	Switch0	Fa0/3
PC3	Switch0	Fa0/4



IP Address Assignment

Configure the following IP addresses on each PC: *(Click each PC → Desktop → IP Configuration)*

Device	VLAN	IP Address	Subnet Mask
PC0	VLAN 10 — Finance	192.168.10.1	255.255.255.0
PC1	VLAN 10 — Finance	192.168.10.2	255.255.255.0
PC2	VLAN 20 — HR	192.168.20.1	255.255.255.0
PC3	VLAN 20 — HR	192.168.20.2	255.255.255.0

Switch Configuration

Click on **Switch0 → CLI** and type the following commands:

Step 1 — Enter privileged mode:

enable ! Enters privileged EXEC mode — gives full access to the switch

configure terminal ! Enters global configuration mode — where changes are made

Step 2 — Create VLAN 10 and name it Finance:

vlan 10 ! Creates VLAN 10 on the switch

name Finance ! Gives VLAN 10 the name Finance — easier to identify

exit ! Exits VLAN configuration mode

Step 3 — Create VLAN 20 and name it HR:

vlan 20 ! Creates VLAN 20 on the switch

name HR ! Gives VLAN 20 the name HR

exit ! Exits VLAN configuration mode

Step 4 — Assign PC0 port to VLAN 10:

interface fa0/1 ! Selects port Fa0/1 where PC0 is connected

switchport mode access ! Sets the port as an access port — carries only one VLAN

switchport access vlan 10 ! Assigns this port to VLAN 10

exit ! Exits interface configuration

Step 5 — Assign PC1 port to VLAN 10:

interface fa0/2 ! Selects port Fa0/2 where PC1 is connected switchport mode access ! Sets the port as an access port

switchport access vlan 10 ! Assigns this port to VLAN 10

exit

Step 6 — Assign PC2 port to VLAN 20:

interface fa0/3 ! Selects port Fa0/3 where PC2 is connected

switchport mode access ! Sets the port as an access port

switchport access vlan 20 ! Assigns this port to VLAN 20

exit

Step 7 — Assign PC3 port to VLAN 20:

interface fa0/4 ! Selects port Fa0/4 where PC3 is connected

switchport mode access ! Sets the port as an access port

switchport access vlan 20 ! Assigns this port to VLAN 20

exit

Step 8 — Verify your configuration:

show vlan brief ! Displays all VLANs and which ports belong to each one

You should see:

- VLAN 10 — Finance — Fa0/1, Fa0/2
- VLAN 20 — HR — Fa0/3, Fa0/4

Testing — What You Need to Do

Test 1 — Same VLAN — Should WORK

- Click **PC0 → Desktop → Command Prompt**
- Type: ping 192.168.10.2
- Expected result: **Ping succeeds** — PC0 and PC1 are in the same VLAN 10

Test 2 — Different VLAN — Should FAIL

- Click **PC0 → Desktop → Command Prompt**
- Type: ping 192.168.20.1
- Expected result: **Ping fails** — PC0 cannot reach PC2 — different VLANs

Test 3 — Same VLAN across HR — Should WORK

- Click **PC2 → Desktop → Command Prompt**
- Type: ping 192.168.20.2
- Expected result: **Ping succeeds** — PC2 and PC3 are in the same VLAN 20

Key Takeaway

PC0 and PC2 are connected to the same physical switch — but they cannot talk to each other. That is the power of VLANs. One switch — two separate worlds.

Classwork Activity 2 — Trunking and 802.1Q Tagging: One Cable, Many VLANs

Objective

By the end of this activity, you will be able to:

- Configure a trunk link between two switches
- Verify that VLANs extend across switches over one cable
- Test communication between same VLAN devices on different switches
- Understand how 802.1Q tagging carries VLAN traffic across a trunk link

Build Your Topology

You will need the following devices:

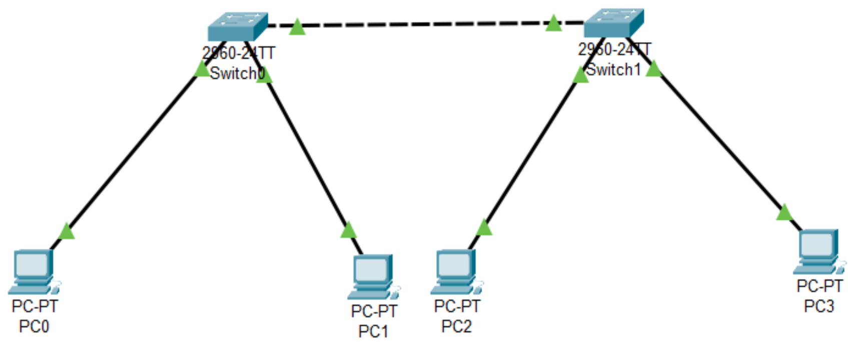
Device	Type	Quantity
Switch	2960	2
PC	PC-PT	4

Connect them as follows:

From	To	Port	Cable
PC0	Switch0	Fa0/1	Straight-through
PC1	Switch0	Fa0/2	Straight-through
PC2	Switch1	Fa0/1	Straight-through
PC3	Switch1	Fa0/2	Straight-through
Switch0	Switch1	Fa0/24 to Fa0/24	Straight-through

Note: The cable between Switch0 and Switch1 on Fa0/24 is your trunk link. This one cable will carry both VLAN 10 and VLAN 20.

Network Diagram



IP Address Assignment

Configure the following IP addresses on each PC. Click each PC, then Desktop, then IP Configuration.

Device	VLAN	IP Address	Subnet Mask	Switch
PC0	VLAN 10 Finance	192.168.10.1	255.255.255.0	Switch0
PC1	VLAN 20 HR	192.168.20.1	255.255.255.0	Switch0
PC2	VLAN 10 Finance	192.168.10.2	255.255.255.0	Switch1
PC3	VLAN 20 HR	192.168.20.2	255.255.255.0	Switch1

Note: PC0 and PC2 are both in VLAN 10 but on different switches. The trunk link is what connects them.

Switch0 Configuration

Click on Switch0, then CLI, and type the following commands.

Step 1 — Enter privileged mode:

enable ! Enter privileged EXEC mode to get full access

configure terminal ! Enter global configuration mode to start making changes

Step 2 — Create VLAN 10 and name it Finance:

vlan 10 ! Create VLAN 10 on this switch

name Finance ! Name it Finance so it is easy to identify

exit ! Exit VLAN configuration and go back one level

Step 3 — Create VLAN 20 and name it HR:

vlan 20 ! Create VLAN 20 on this switch

name HR ! Name it HR so it is easy to identify

exit ! Exit VLAN configuration and go back one level

Step 4 — Assign PC0 to VLAN 10:

interface fa0/1 ! Select the port where PC0 is connected

switchport mode access ! Set this port to carry only one VLAN

switchport access vlan 10 ! Assign this port to VLAN 10

exit ! Exit interface configuration

Step 5 — Assign PC1 to VLAN 20:

interface fa0/2 ! Select the port where PC1 is connected

switchport mode access ! Set this port to carry only one VLAN

switchport access vlan 20 ! Assign this port to VLAN 20

exit ! Exit interface configuration

Step 6 — Configure the trunk port toward Switch1:

interface fa0/24 ! Select the port connected to Switch1

switchport mode trunk ! Set this port to carry all VLANs at the same time

exit ! Exit interface configuration

Step 7 — Verify Switch0 configuration:

show vlan brief ! Display all VLANs and their assigned ports

show interfaces trunk ! Display which ports are trunks and which VLANs they carry

Switch1 Configuration

Click on Switch1, then CLI, and type the following commands.

Step 1 — Enter privileged mode:

enable ! Enter privileged EXEC mode to get full access

configure terminal ! Enter global configuration mode to start making changes

Step 2 — Create VLAN 10 and name it Finance:

vlan 10 ! Create VLAN 10 on this switch — must match Switch0

name Finance ! Name it Finance to match Switch0

exit ! Exit VLAN configuration and go back one level

Step 3 — Create VLAN 20 and name it HR:

vlan 20 ! Create VLAN 20 on this switch — must match Switch0

name HR ! Name it HR to match Switch0

exit ! Exit VLAN configuration and go back one level

Step 4 — Assign PC2 to VLAN 10:

interface fa0/1 ! Select the port where PC2 is connected

switchport mode access ! Set this port to carry only one VLAN

switchport access vlan 10 ! Assign this port to VLAN 10

exit ! Exit interface configuration

Step 5 — Assign PC3 to VLAN 20:

interface fa0/2 ! Select the port where PC3 is connected

switchport mode access ! Set this port to carry only one VLAN

switchport access vlan 20 ! Assign this port to VLAN 20

exit ! Exit interface configuration

Step 6 — Configure the trunk port toward Switch0:

interface fa0/24 ! Select the port connected to Switch0

switchport mode trunk ! Set this port to carry all VLANs at the same time

exit ! Exit interface configuration

Step 7 — Verify Switch1 configuration:

show vlan brief ! Display all VLANs and their assigned ports

show interfaces trunk ! Confirm Fa0/24 is operating as a trunk port

Testing — What You Need to Do

Test 1 — Same VLAN across switches — Should WORK

- Click PC0, then Desktop, then Command Prompt
- Type: ping 192.168.10.2
- Expected result: Ping succeeds — PC0 and PC2 are both in VLAN 10 — trunk carries the traffic

Test 2 — Different VLAN — Should FAIL

- Click PC0, then Desktop, then Command Prompt
- Type: ping 192.168.20.1

- Expected result: Ping fails — PC0 is VLAN 10 — PC1 is VLAN 20 — different VLANs cannot communicate

Test 3 — Same VLAN across switches for HR — Should WORK

- Click PC1, then Desktop, then Command Prompt
- Type: ping 192.168.20.2
- Expected result: Ping succeeds — PC1 and PC3 are both in VLAN 20 — trunk carries the traffic

What Is Happening With 802.1Q — Understand This

When PC0 sends traffic to PC2 — here is exactly what happens step by step:

Step	What Happens
1	PC0 sends a normal frame — no tag
2	Switch0 receives it on Fa0/1 — access port for VLAN 10
3	Switch0 adds an 802.1Q tag — VLAN ID = 10
4	Tagged frame travels across Fa0/24 — the trunk port
5	Switch1 receives the tagged frame — reads the VLAN ID
6	Switch1 removes the tag — delivers frame to PC2 on Fa0/1
7	PC2 receives a normal frame — no tag visible

Note: The tag is added and removed automatically by the switches. The PCs never see it. It only exists on the trunk link between the two switches.

Key Takeaway

One cable connects two switches. That one cable carries VLAN 10 and VLAN 20 at the same time. Without the trunk link — PC0 and PC2 could never communicate even though they are in the same VLAN. 802.1Q tagging is what makes this possible.

Bonus Activity — Enterprise Network Segmentation

Cisco Packet Tracer — Advanced Hands-On Exercise

Scenario

You have just been hired as a junior network administrator at a small company called SecureNet Inc. The company has two floors. Each floor has its own switch. The company has three departments — Sales, IT, and Management. Your job is to segment the network so that each department is isolated from the others, while still allowing devices in the same department to communicate across both floors.

Your Mission

- Build the network from scratch
- Assign each department its own VLAN
- Make sure devices in the same department can talk to each other even if they are on different floors
- Make sure devices in different departments cannot talk to each other
- Verify everything works using ping tests

Device List — Build This Topology

Device	Type	Quantity
Switch	2960	2
PC	PC-PT	6

Connection Table — How to Wire Everything

Device	Connected To	Port	Cable Type
PC0 — Sales Staff 1	Switch0 — Floor 1	Fa0/1	Straight-through
PC1 — IT Staff 1	Switch0 — Floor 1	Fa0/2	Straight-through
PC2 — Management 1	Switch0 — Floor 1	Fa0/3	Straight-through
PC3 — Sales Staff 2	Switch1 — Floor 2	Fa0/1	Straight-through
PC4 — IT Staff 2	Switch1 — Floor 2	Fa0/2	Straight-through
PC5 — Management 2	Switch1 — Floor 2	Fa0/3	Straight-through
Switch0 — Floor 1	Switch1 — Floor 2	Fa0/24 to Fa0/24	Straight-through

VLAN Assignment Table

VLAN ID	Department	Color Code
VLAN 10	Sales	Red
VLAN 20	IT	Blue
VLAN 30	Management	Green

IP Address Assignment

Device	Department	VLAN	IP Address	Subnet Mask	Floor
PC0	Sales Staff 1	VLAN 10	192.168.10.1	255.255.255.0	Floor 1
PC1	IT Staff 1	VLAN 20	192.168.20.1	255.255.255.0	Floor 1
PC2	Management 1	VLAN 30	192.168.30.1	255.255.255.0	Floor 1
PC3	Sales Staff 2	VLAN 10	192.168.10.2	255.255.255.0	Floor 2
PC4	IT Staff 2	VLAN 20	192.168.20.2	255.255.255.0	Floor 2
PC5	Management 2	VLAN 30	192.168.30.2	255.255.255.0	Floor 2

Your Configuration Tasks

Task 1 — Configure Switch0 (Floor 1)

- Create all three VLANs with their correct names
- Assign each PC port to its correct VLAN
- Configure the uplink port Fa0/24 as a trunk toward Switch1

Task 2 — Configure Switch1 (Floor 2)

- Create all three VLANs with their correct names — they must match Switch0
- Assign each PC port to its correct VLAN
- Configure the uplink port Fa0/24 as a trunk toward Switch0

Task 3 — Verify Your Configuration

- Run show vlan brief on both switches
- Run show interfaces trunk on both switches
- Confirm all VLANs appear correctly on both switches
- Confirm Fa0/24 is listed as a trunk port on both switches

Testing — Ping Tests You Must Complete

Complete all six ping tests and record whether each one succeeded or failed.

Test	From	To	IP Address	Expected Result	Your Result
1	PC0 Sales 1	PC3 Sales 2	192.168.10.2	Should WORK	
2	PC1 IT 1	PC4 IT 2	192.168.20.2	Should WORK	
3	PC2 Management 1	PC5 Management 2	192.168.30.2	Should WORK	

4	PC0 Sales 1	PC1 IT 1	192.168.20.1	Should FAIL	
5	PC1 IT 1	PC2 Management 1	192.168.30.1	Should FAIL	
6	PC0 Sales 1	PC5 Management 2	192.168.30.2	Should FAIL	

Hints — If You Are Stuck

- **Hint 1**: The trunk configuration is exactly the same as the previous activity — only the number of VLANs is different
- **Hint 2**: Each department needs its own VLAN — three departments means three VLANs
- **Hint 3**: The IP address of each PC must match its VLAN subnet — Sales uses 192.168.10.x, IT uses 192.168.20.x, Management uses 192.168.30.x
- **Hint 4**: Run show vlan brief after configuration — if a port is missing from its VLAN, you may have skipped an assignment step
- **Hint 5**: If a ping that should work is failing — check that both switches have the same VLAN names and IDs

Submission Checklist

Before you submit — confirm you have completed the following:

- All six PCs have correct IP addresses configured
- All three VLANs are created on both switches with correct names
- All PC ports are assigned to the correct VLANs on both switches
- Fa0/24 is configured as trunk on both switches
- All six ping tests are completed with results recorded
- All six reflection questions are answered